

A Resilient On-Chip STDP Learning Under Memory Voltage Scaling

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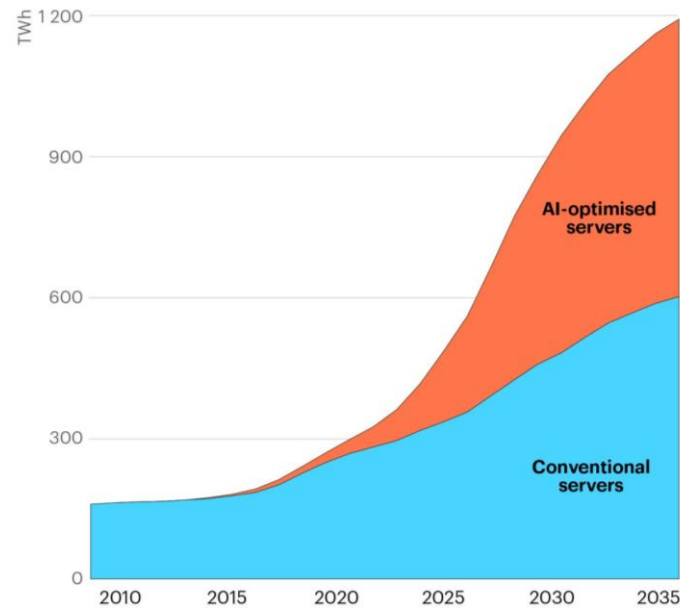
Content

- Introduction
- Proposed Method
- Results
- Conclusion

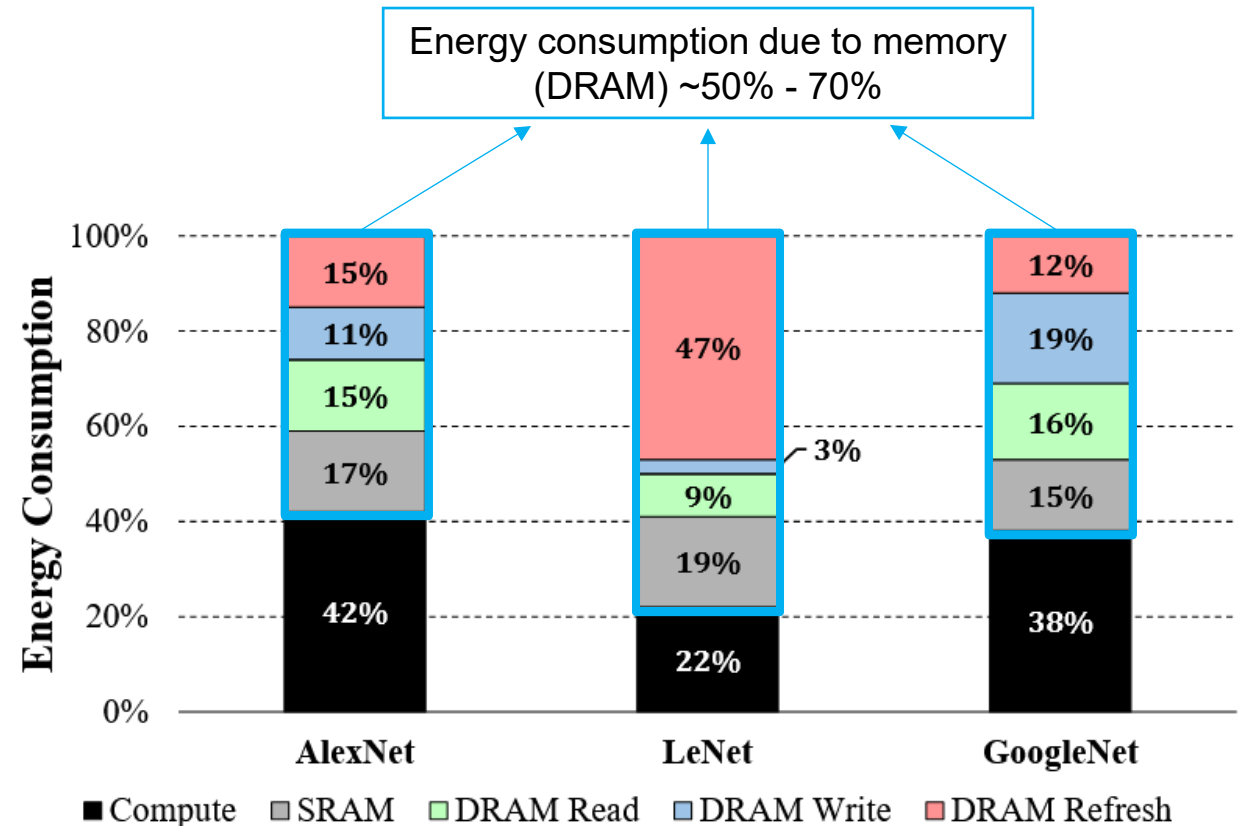
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 - Spiking Neural Networks (SNNs)
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 - On-Chip STDP Learning
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 - Voltage effects on DRAM and SNNs
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Introduction

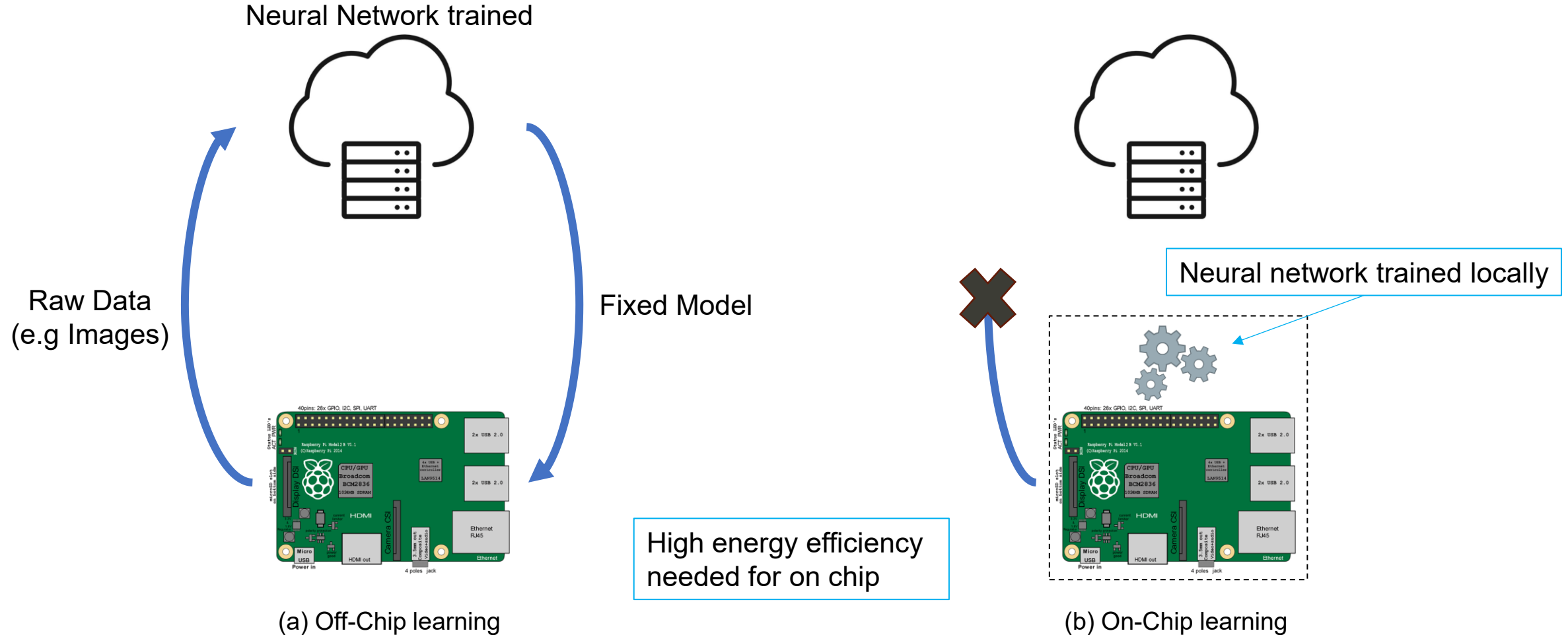


(a) Predicted power consumption due to AI usage ^[1]

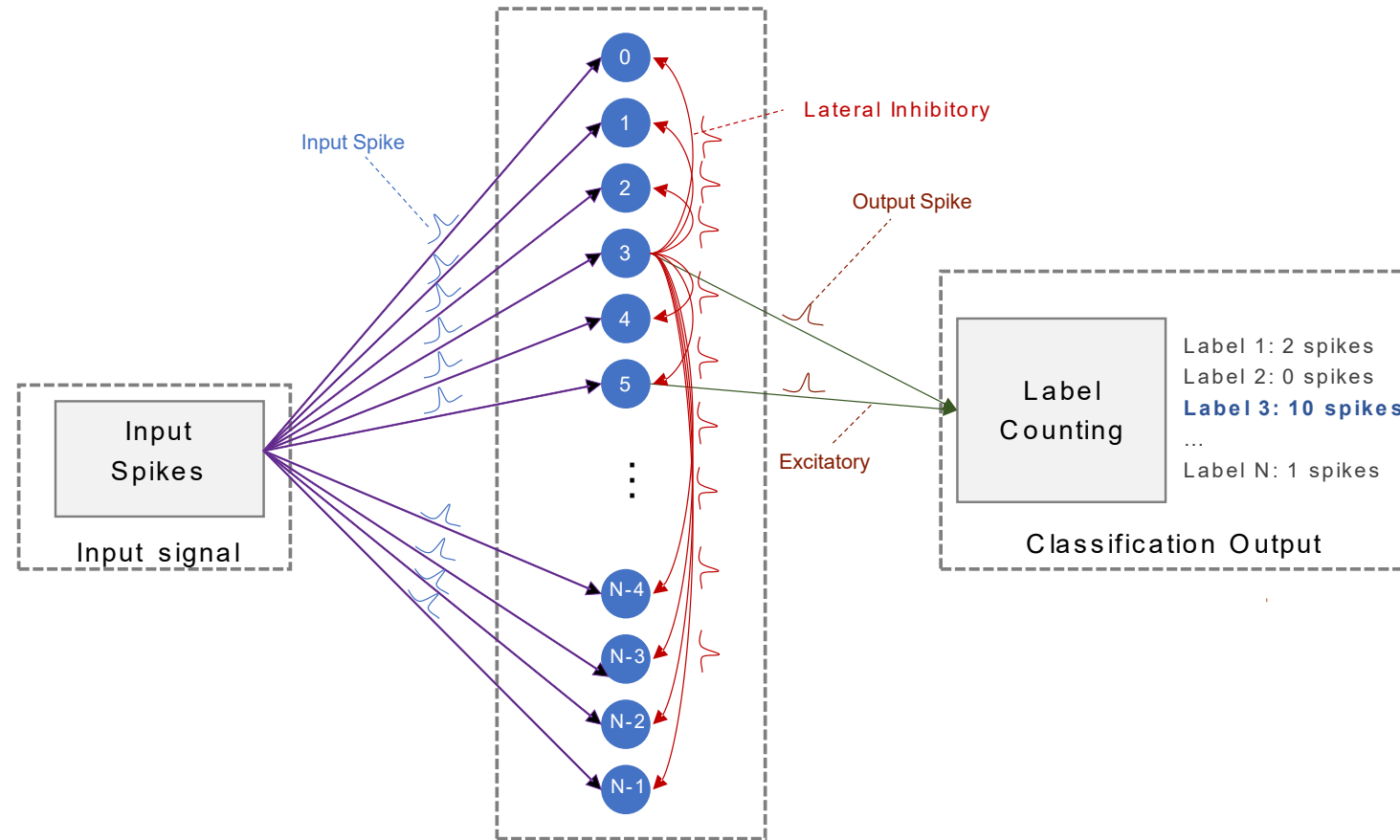


(b) Energy consumption breakdown of three CNNs on a modern CNN accelerator ^[2]

Off-Chip and On-Chip Learning



Spiking Neural Networks

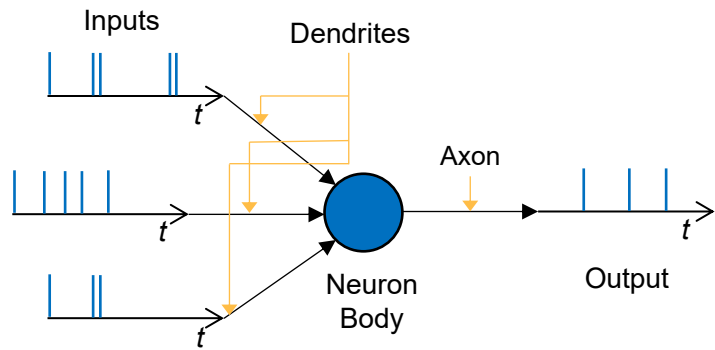


A Spiking Neural Network (SNN) topology employing On-Chip Spike-Timing Dependent Plasticity (STDP)

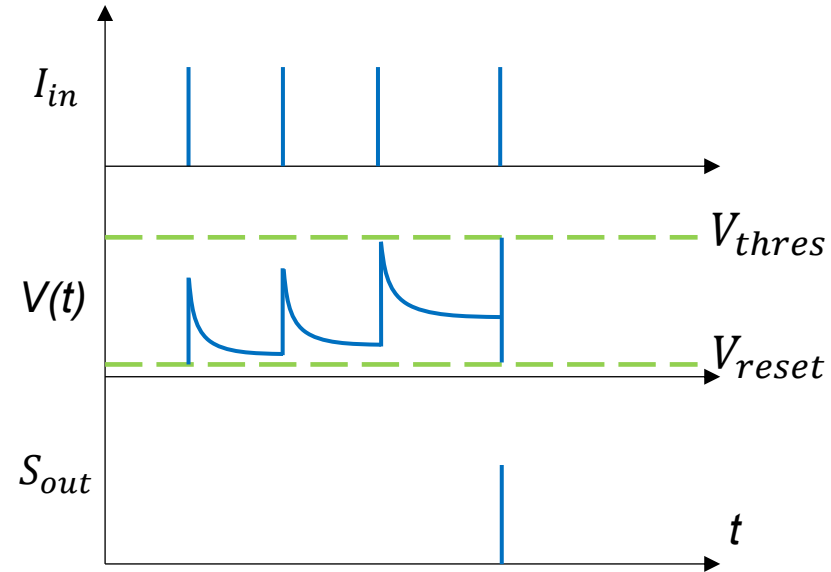
Leaky Integrate and Fire (LIF) Neuron

τ : Time constant
 V : Membrane potential
 E_{rest} : Resting potential
 E_{exc} : Excitatory potential
 E_{inh} : Inhibitory potential
 g_i, g_e : Dynamic Conductance
 I_{in} : Input current

S_{out} : Output spike
 V_{thres} : Threshold potential
 V_{reset} : Reset potential



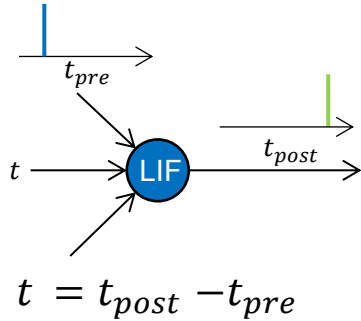
(a) Typical SNN Neuron



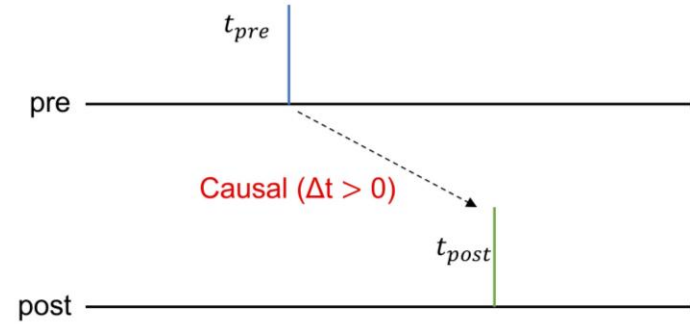
(b) LIF Potential dynamics

$$\text{LIF equation: } \tau \frac{dV}{dt} = (E_{rest} - V) + g_e(E_{exc} - V) + g_i(E_{inh} - V)$$

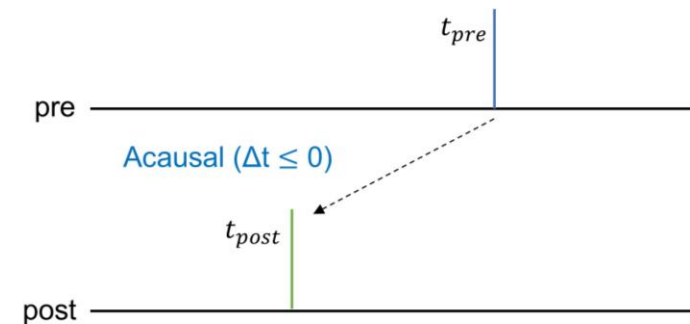
On-Chip STDP Learning



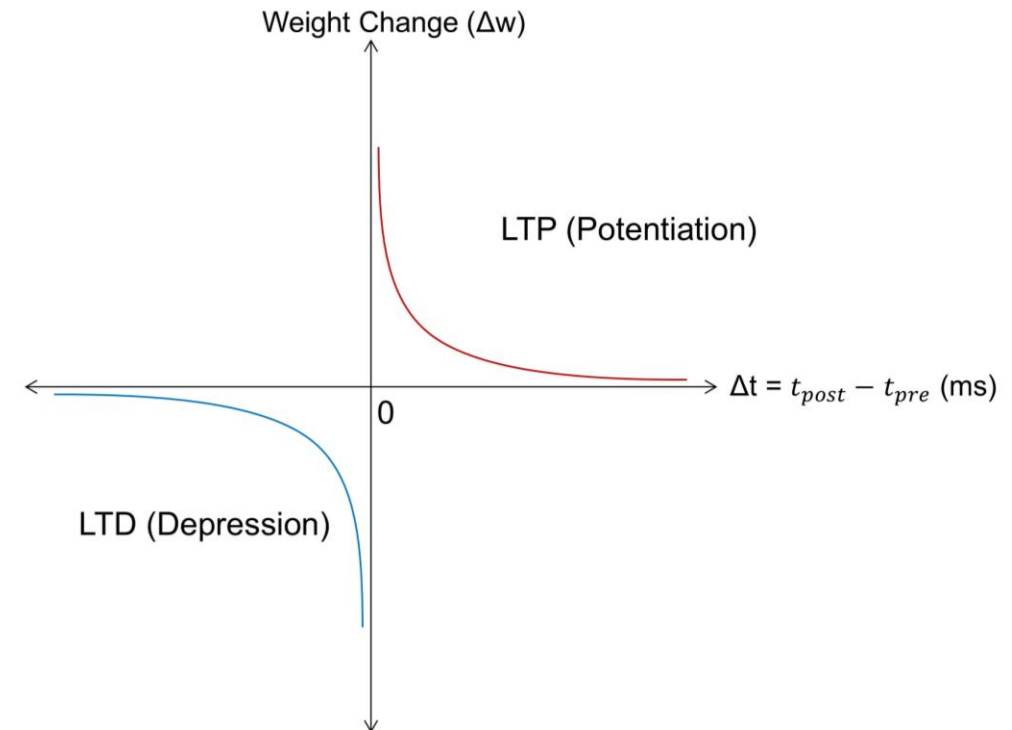
(a) LIF neuron and spike timings



(b) LTP Timing



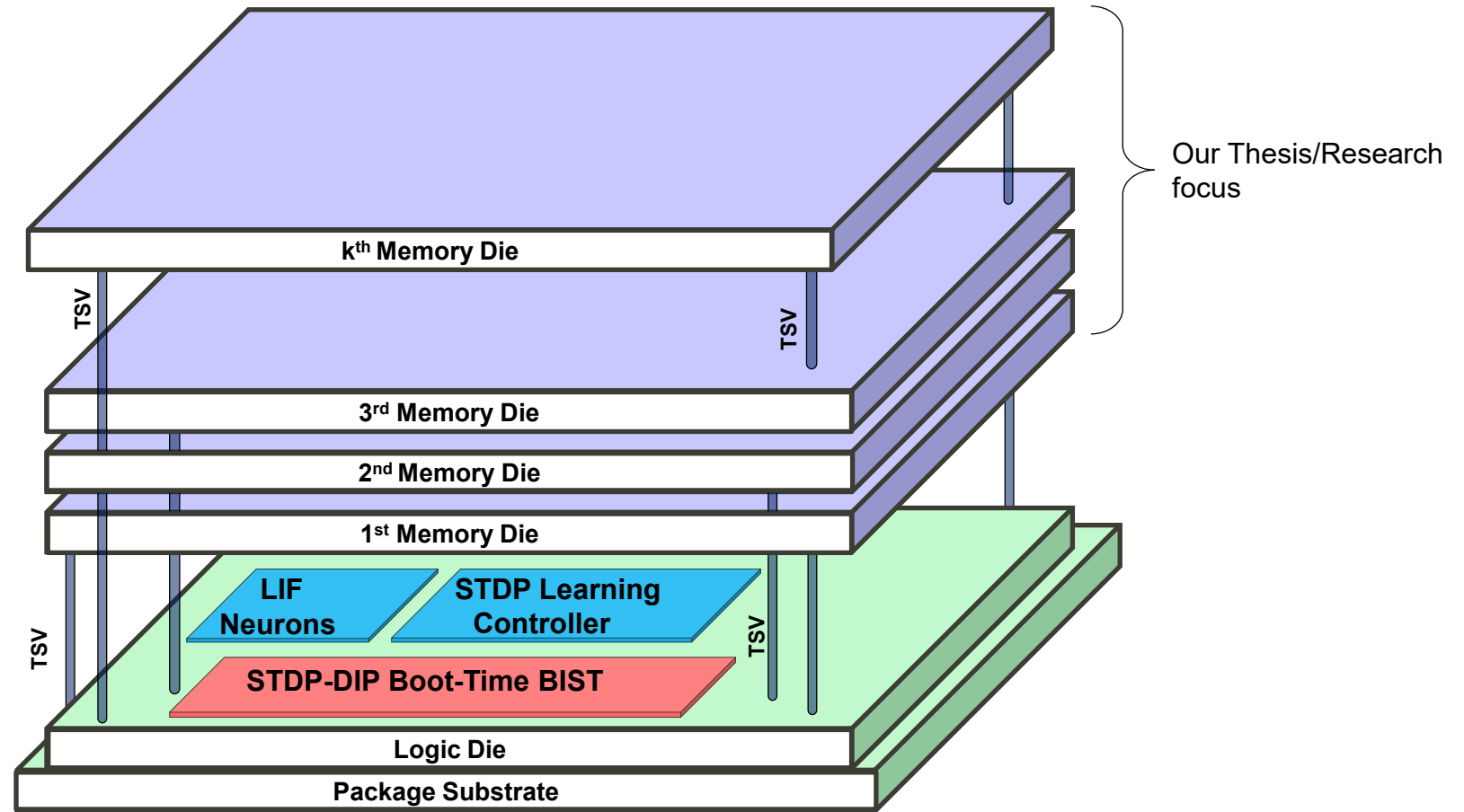
(c) LTD Timing



(d) STDP Learning Window

$$\text{Equation: } \Delta w = \eta (x_{pre} - x_{tar}) (w_{max} - w)^\mu$$

Overall Architecture

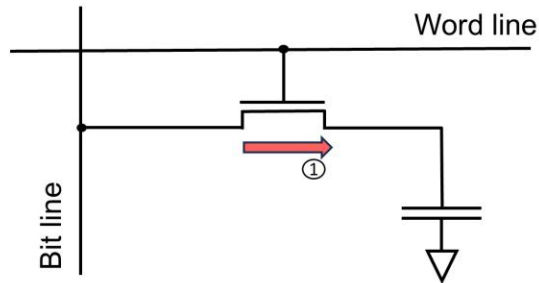


Architecture of proposed system

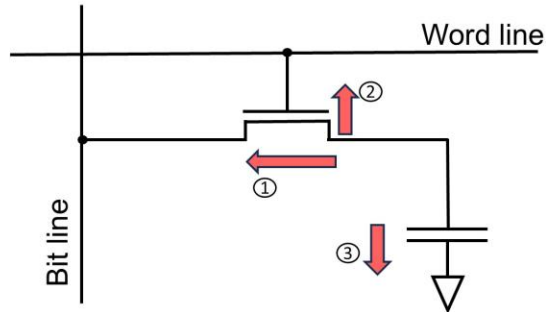
Part of "Y. Hanyu, R. Kobayashi, S. R. Hariprakash, S. Ganesh, N.-D. Nguyen, D.-A. Nguyen, M. Maatar, A. Ben Abdallah, Z. Wang, and K. N. Dang, 'E3STDP: Energy-Efficient Ensemble Spike-Timing-Dependent Plasticity for Carbon-Aware Neuromorphic Computation.'" IEEE Transactions on Sustainable Computing (*under major revision*)

Undervolting DRAM

① Subthreshold leakage ② Gate-induced drain leakage ③ Junction leakage

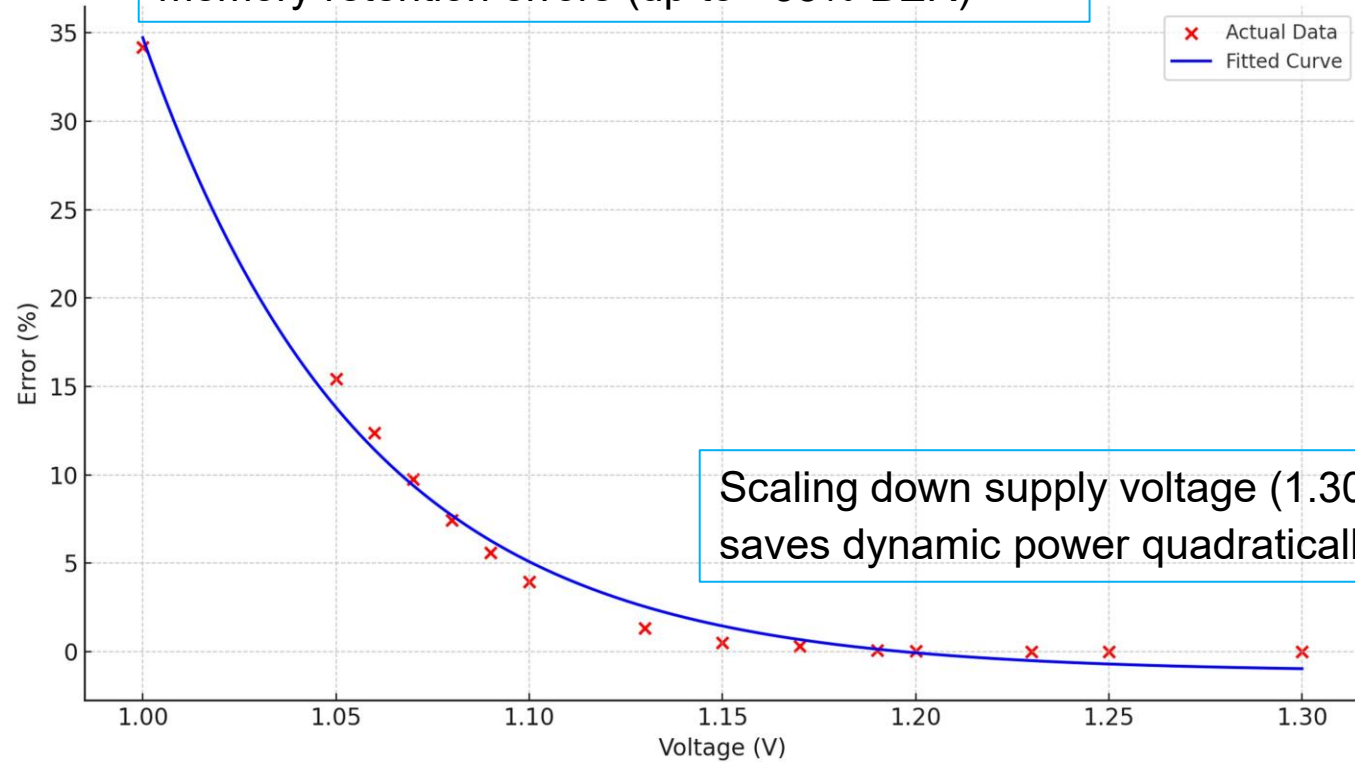


(a) Incoming Leakage path



(b) Outgoing Leakage paths

Trade-off: Triggers an exponential increase in memory retention errors (up to ~35% BER)



Scaling down supply voltage (1.30V to 1.00V) saves dynamic power quadratically ($P \propto V_{DD}^2$)

(c) Graph showing Voltage vs Error obtained from HSPICE simulations using a 45 nm CMOS technology

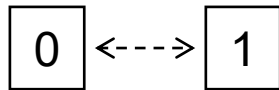
Memory Fault Behaviours Under Undervolting

Soft Errors (Bit-Flips):

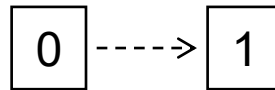
- Temporary data glitches

Permanent Errors:

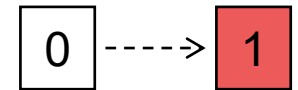
- Caused by manufacturing defects
- Exposed due to aggressive undervolting
- This research focuses on Stuck-at Faults
- Weak cells leak charge, permanently locks physical cell to ground (0) or supply voltage (1)



(a) Normal bit update

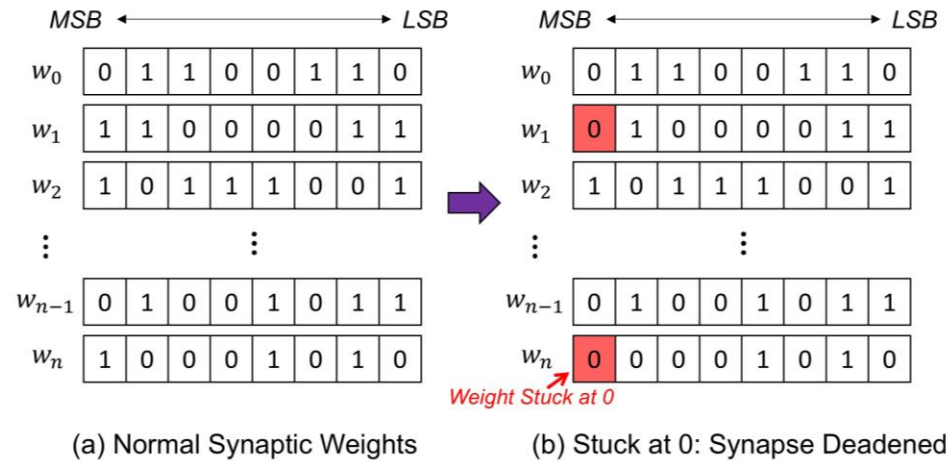


(b) Bit flip



(c) Bit stuck-at

Impact of SA0 vs SA1 Faults

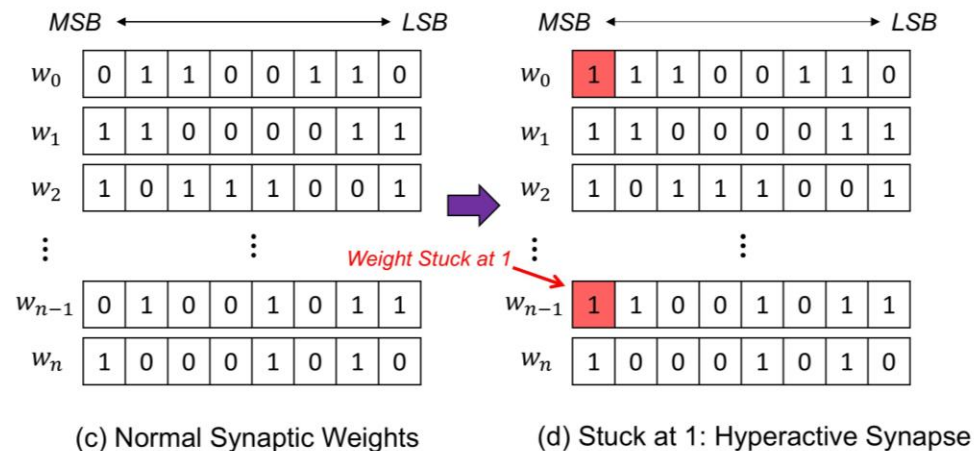


Stuck-at-0 (SA0):

- Weight locked to minimum conductance
- Easily absorbed, accuracy remains stable (>90%).

Stuck-at-1 (SA1):

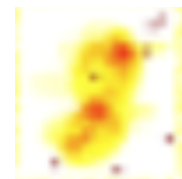
- Weight locked to maximum conductance
- Neuron fires continuously
- Suppresses all healthy neurons
- Accuracy collapses from **92.0% to 10.9%***



(e) Normal Field



(f) SA0 Field



(g) SA1 Field

*Data obtained from STDP Learning model with 784 weights per 1000 neuron network, MNIST dataset

Related Work

- Error Correction Codes ^[3]:
 - Requires 12.5% to 37.5% storage area overhead
 - Encoding/decoding negates 40% energy saved
- Checkpointing ^[4] :
 - Milliseconds of latency per save
 - Incurs >100% to 200% power overhead
- Inference Mitigations ^[5] :
 - Protects only pre-trained networks
 - Energy overhead 15% to 30%
- Memory Checking ^[6] :
 - Can check for MSB and LSB
 - Energy overhead 30% to 50%

Unsuitable for On-chip STDP learning

Problem Statement and Contributions

Problem:

- Near-threshold undervolting to lower consumption
- Low voltages cause stuck-faults and flip bits
- SA1 destroys STDP learning

Contributions:

- Characterized SA1 behavior under eDRAM voltage scaling
- Proposed **STDP-DIP** to identify, isolate and prune faulty neurons

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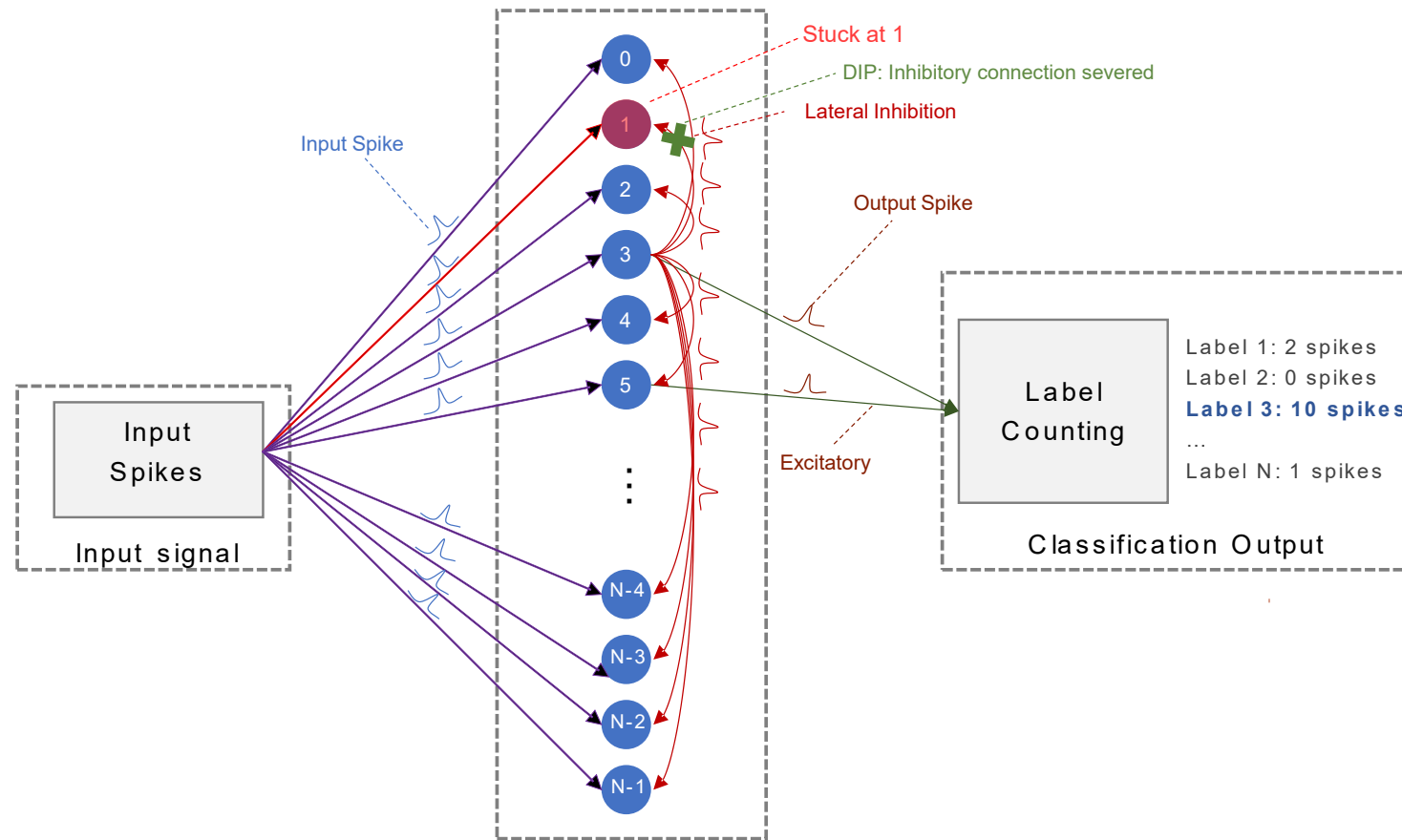
STDP-DIP: Diagnostic Isolation Pruning

Phase 1: Detects SA1 at MSB in the weight matrix

Phase 2: Faulty neuron is structurally severed

Phase 3: Output of faulty neuron unlabeled

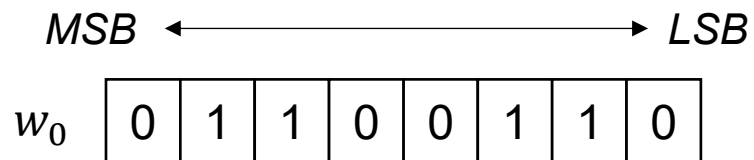
STDP-DIP: Proposal Overview



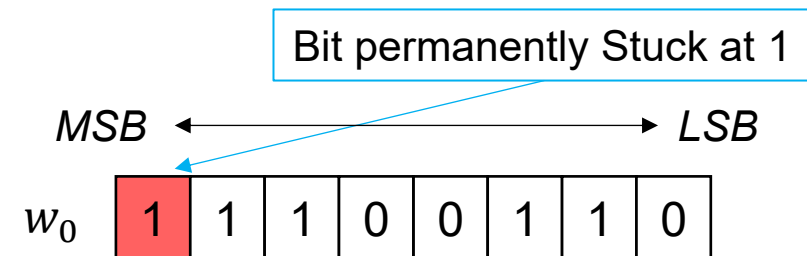
Overview of STDP network and STDP-DIP

STDP-DIP: Detection

- MSB faults observed to cause most impact
- Synaptic weights of neurons checked for Stuck-at-1 MSBs
- Another way is to detect hyperactive (fires continuously) neuron
- Why?
 - Memory check exists but needs 30% to 50% more energy
 - ECC negates 40% energy saved

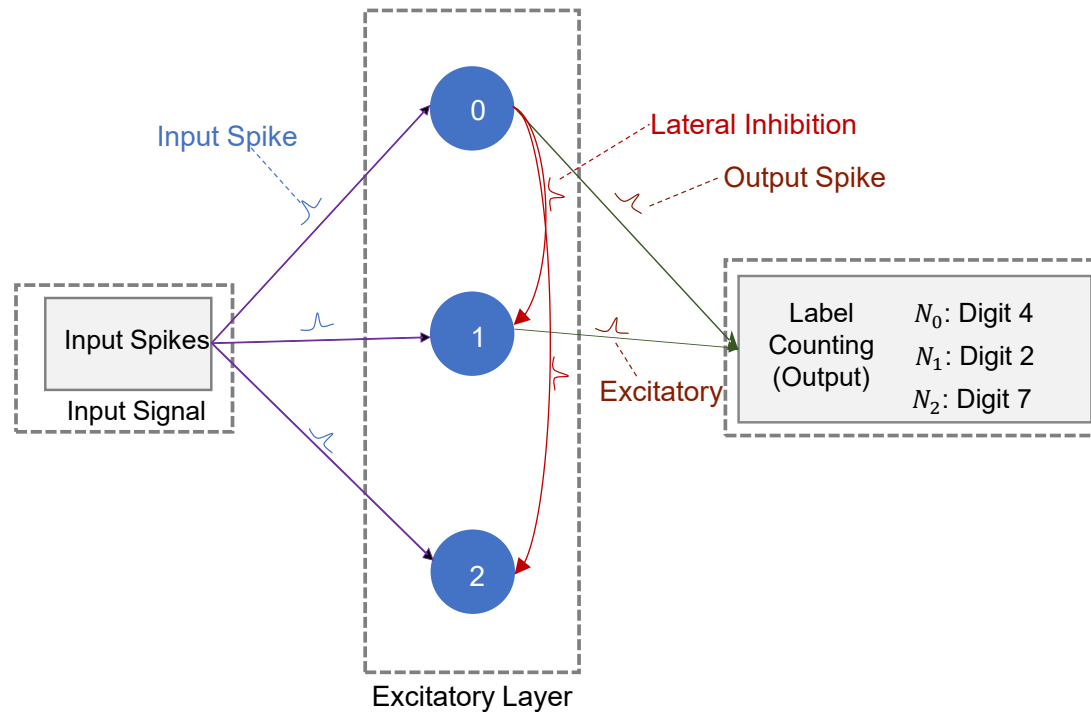


(a) Bit values of synaptic weights of healthy neuron



(b) Bit values of synaptic weights of SA1 neuron

STDP-DIP: Algorithm (1)

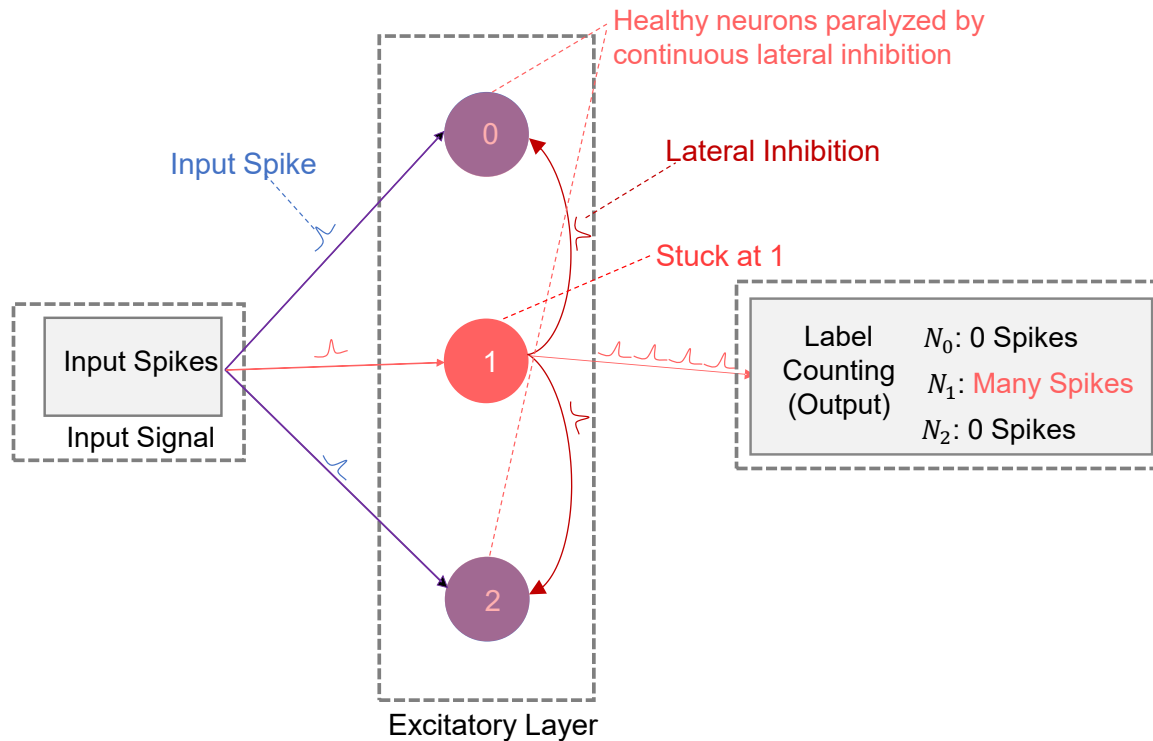


Normal STDP Operation

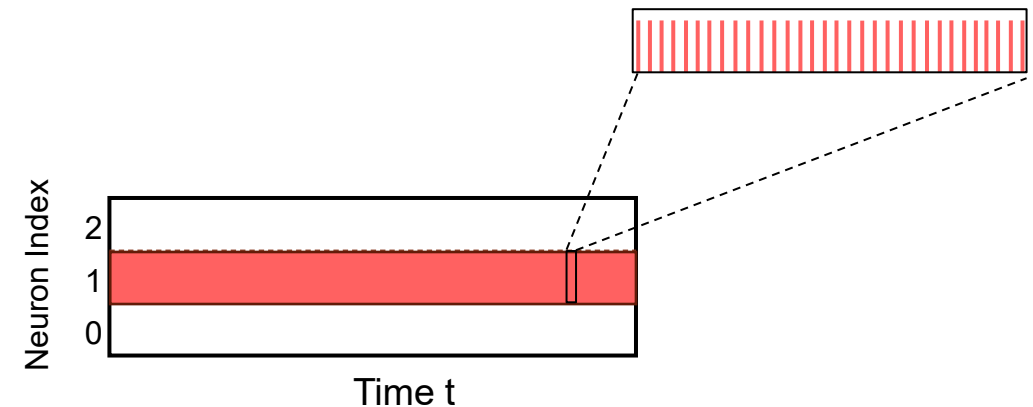
- Brief lateral inhibition triggered by each spike
- Healthy competition and diverse feature learning

STDP-DIP: Algorithm (2)

- Faulty neuron fires continuously
- Runaway lateral inhibition, paralyzes neurons, stops learning

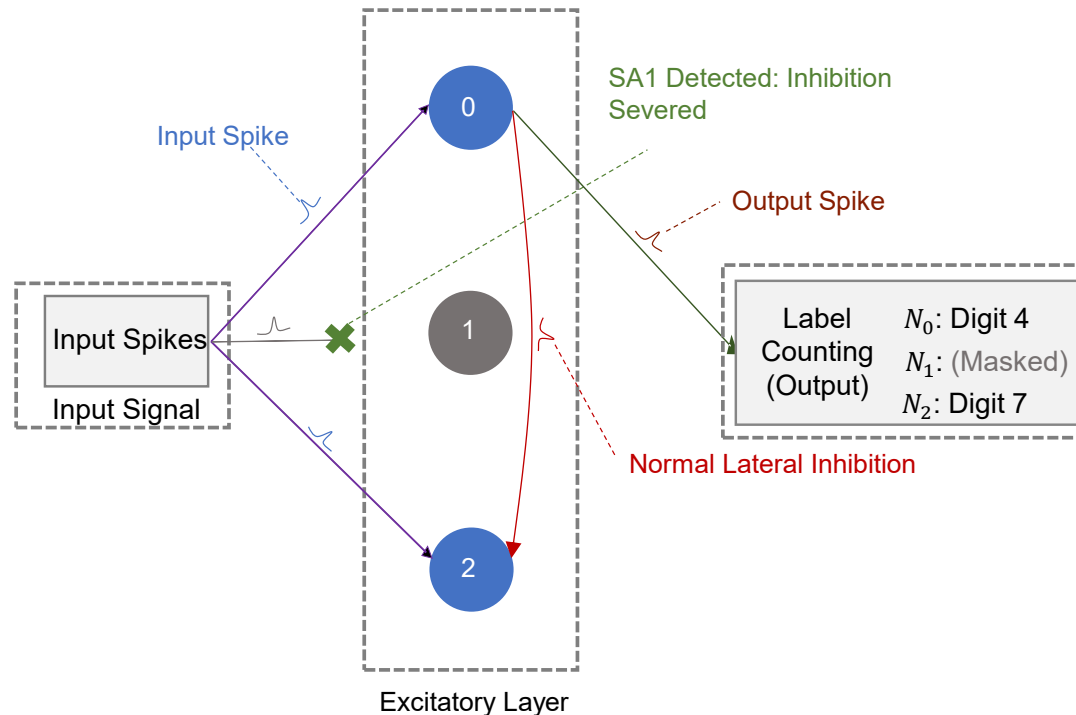


(a) Uncorrected SA1 Fault



(b) Spike timing

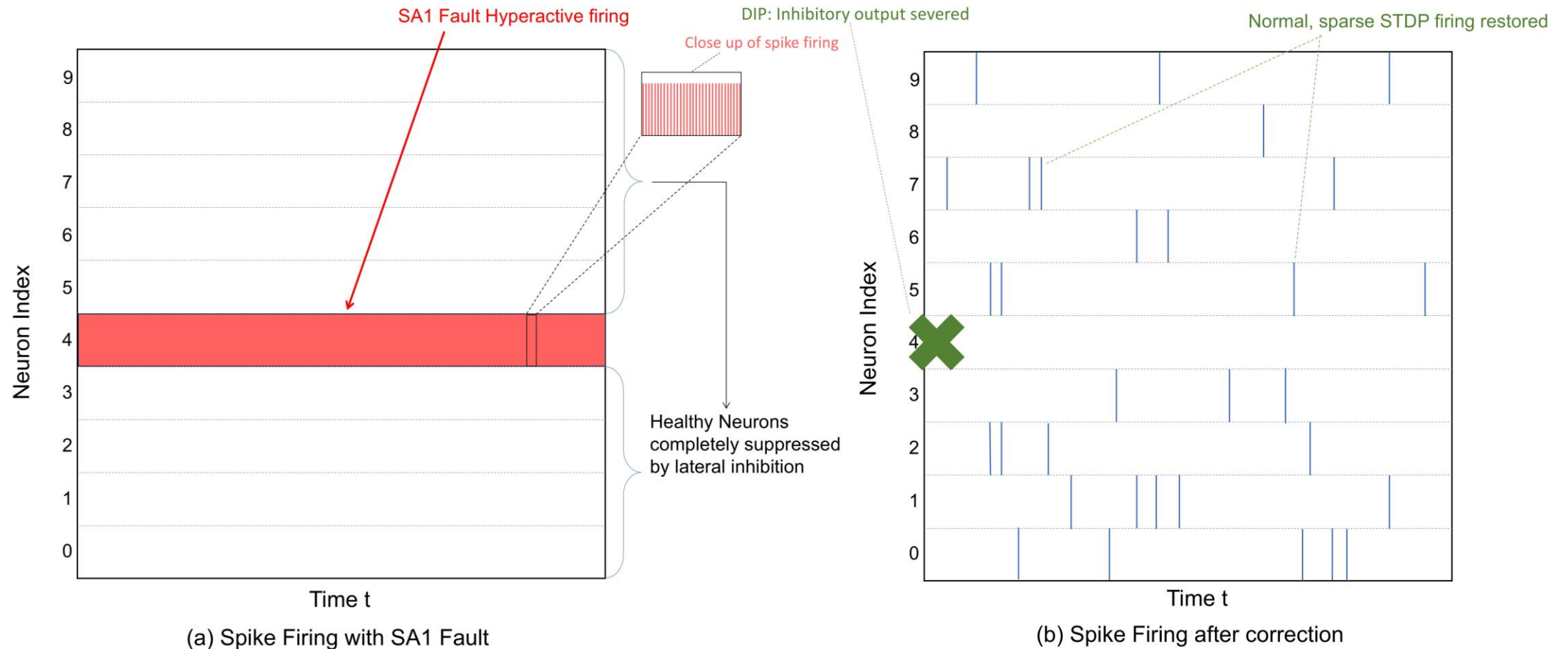
STDP-DIP: Algorithm (3)



Corrected using STDP-DIP

- Detect faulty neuron
- Faulty neuron connection severed
- Healthy STDP spiking restored

Spike Behaviour Restoration



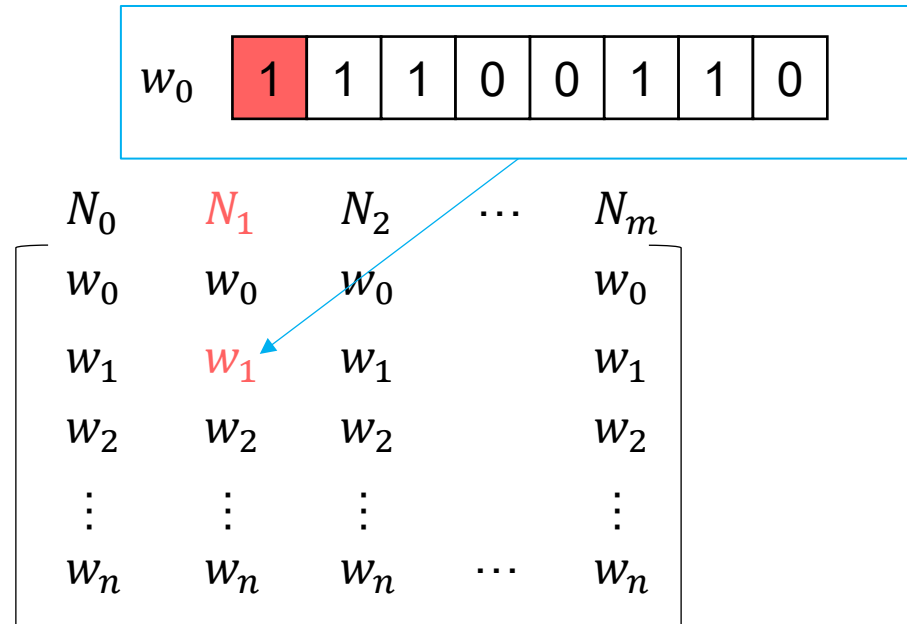
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Simulation Setup

- Environment: Python, PyTorch, BindsNET, Ran with 3 different seeds
- Hardware Model: 45nm PTM CMOS eDRAM, Monte Carlo simulation done with HSPICE (100000 trials per case)
- Network: LIF-STDP model
- Dataset: Poisson-encoded MNIST

Fault Insertion

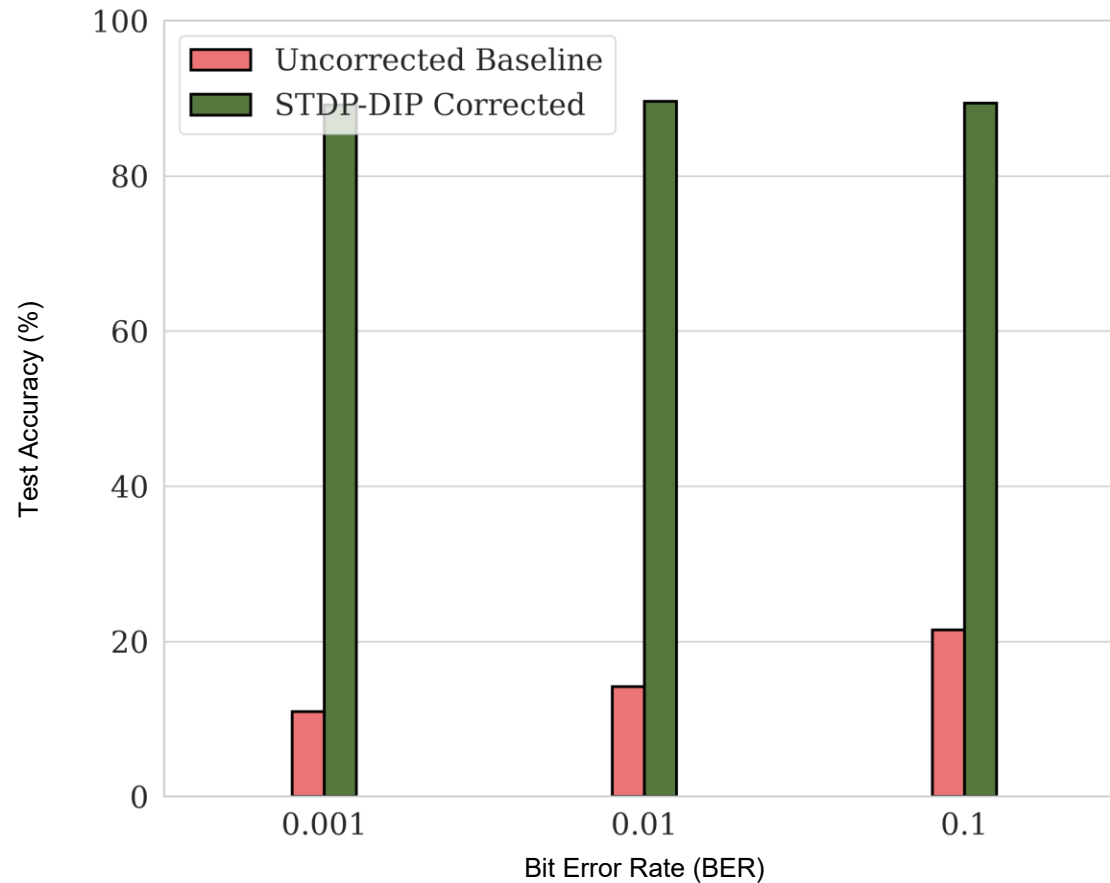


- Faulty neurons chosen randomly
- MSB of weights flipped manually
- Simulates SA1 fault

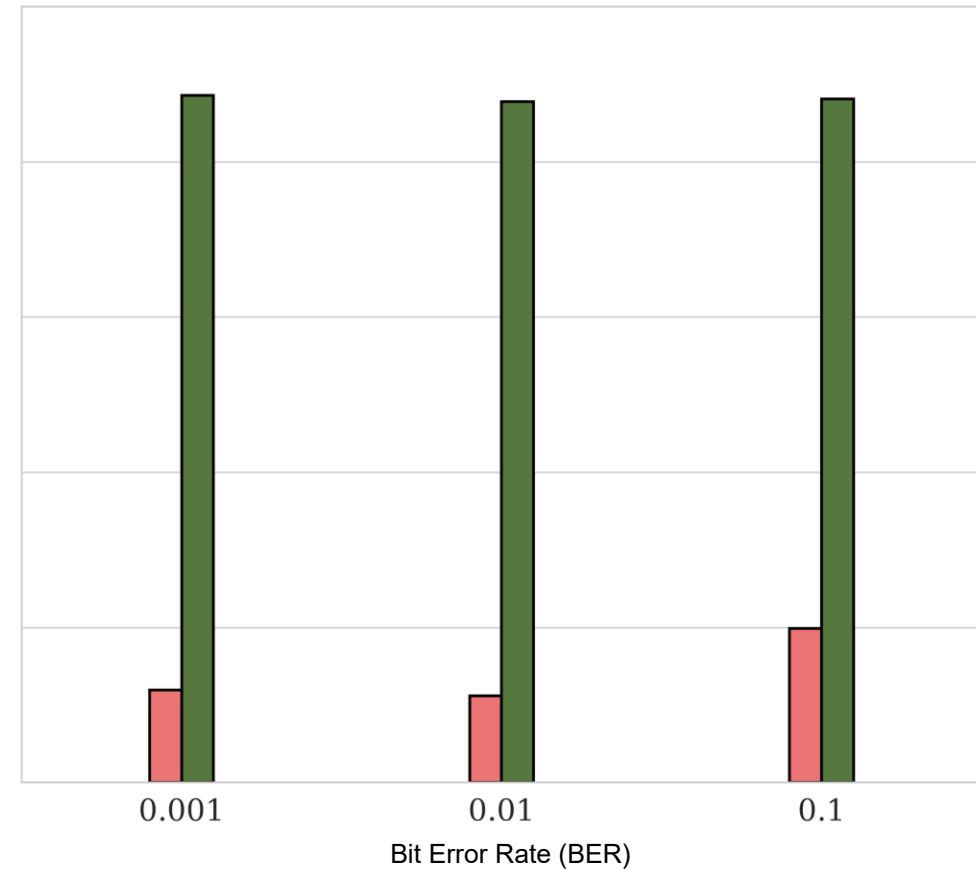
m : Number of Neurons (1000)

n : Number of weights (784)

Results: 400 Neurons



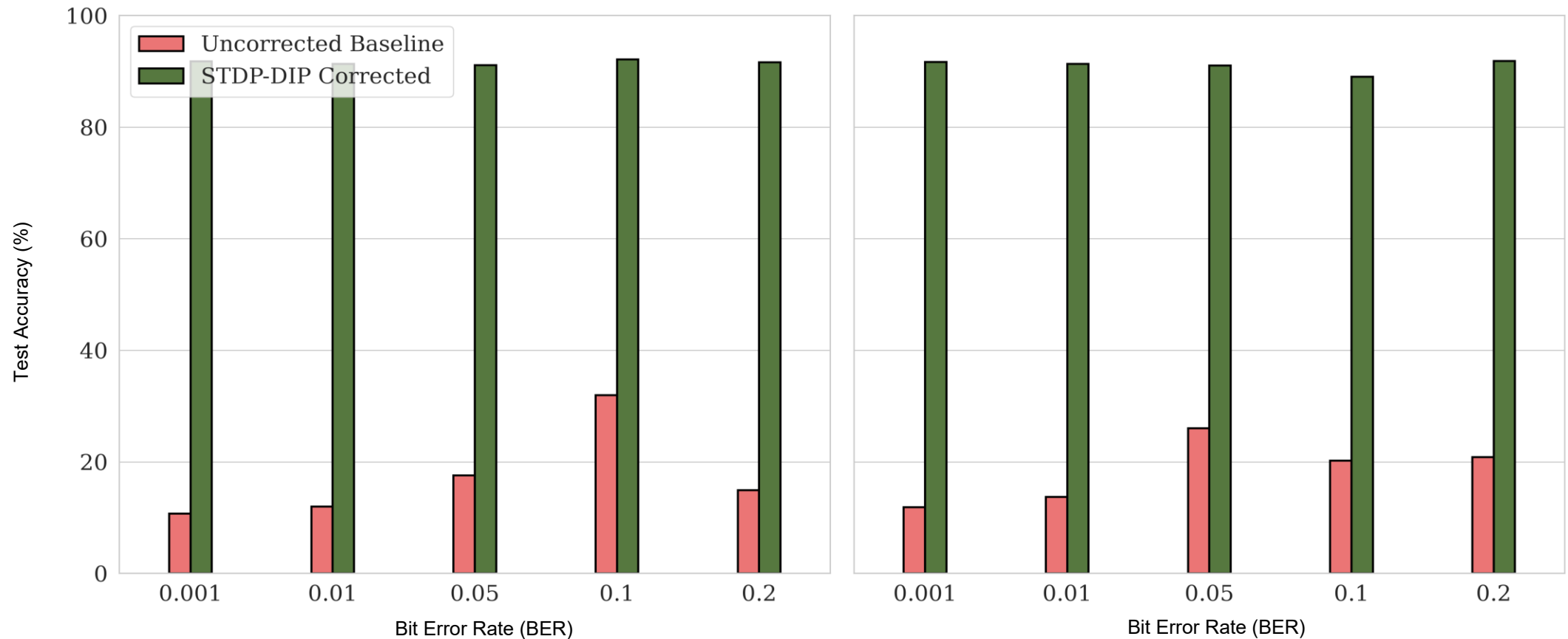
(a) Plot between accuracy before and after correction for 5 SA1 neurons in a 400 neuron network



(b) Plot between accuracy before and after correction for 15 SA1 neurons in a 400 neuron network

Accuracy improved by **73% - 79%**

Results: 1000 Neurons



(a) Plot between accuracy before and after correction for 5 SA1 neurons in a 1000 neuron network

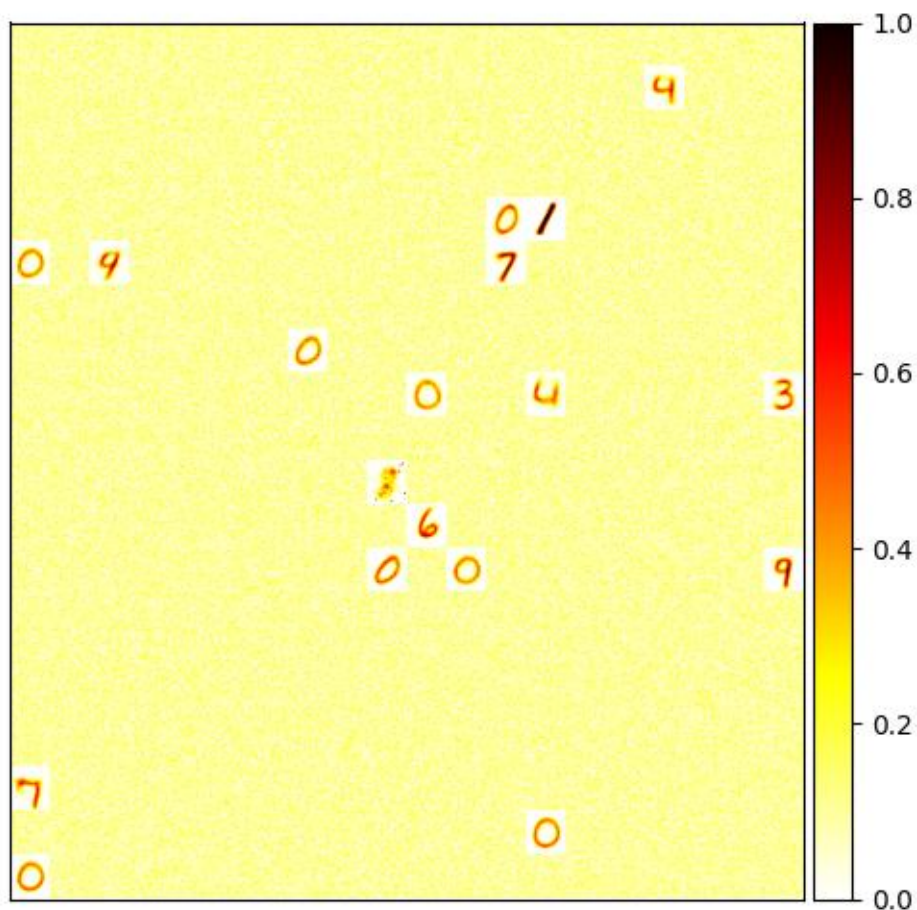
(b) Plot between accuracy before and after correction for 15 SA1 neurons in a 1000 neuron network

Accuracy improved by 71% - 81%

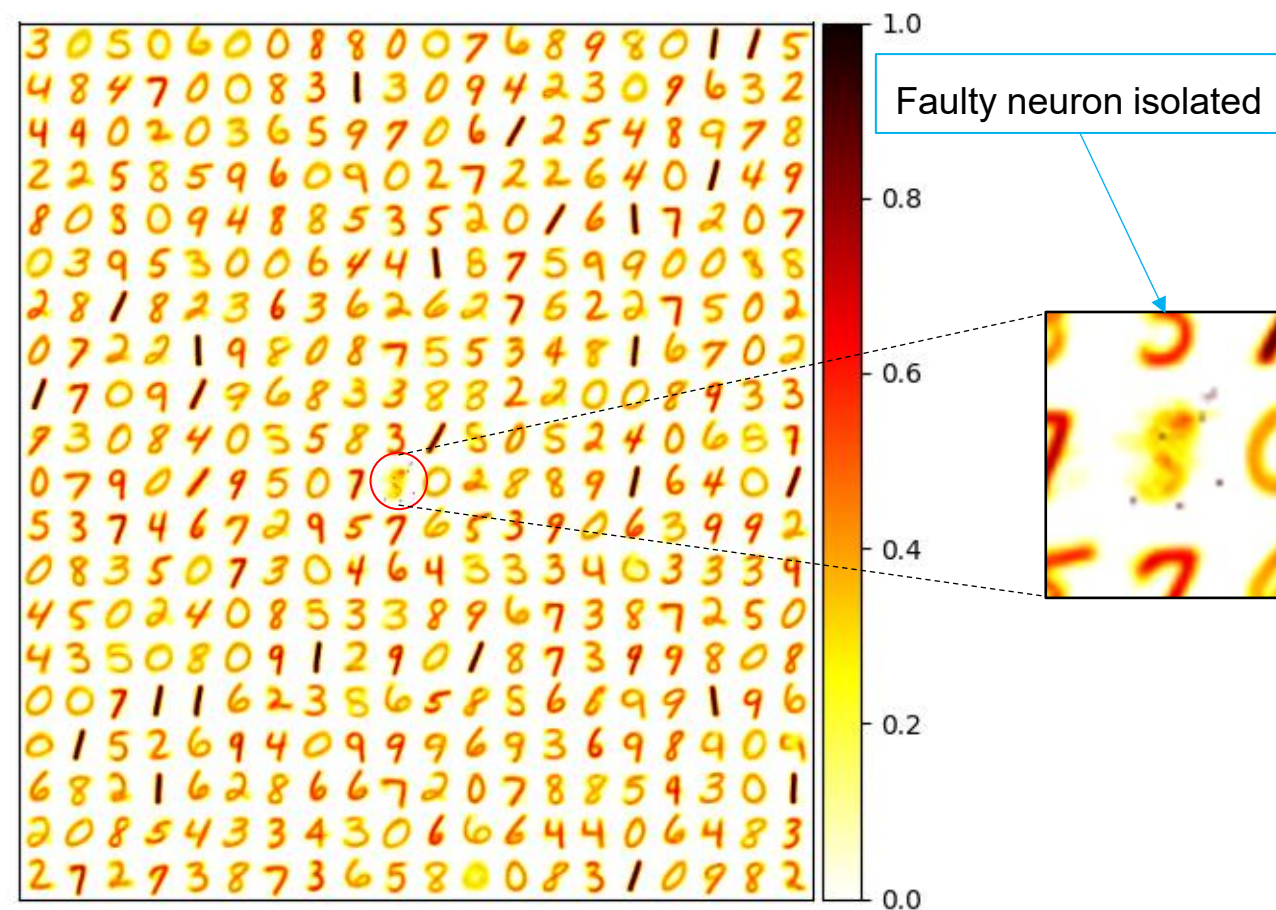
Results: All cases

Neurons	BER (β)	Faulty Neurons	Uncorrected Accuracy (%)	Corrected Accuracy (%)
100	0.001	5	12.18 ± 1.77	77.31 ± 1.03
100	0.01	5	10.05 ± 0.26	77.10 ± 1.35
100	0.1	5	16.97 ± 0.81	77.31 ± 0.44
400	0.001	5	10.94 ± 1.25	89.13 ± 0.28
400	0.01	5	14.14 ± 2.30	89.62 ± 0.22
400	0.1	5	21.47 ± 0.16	89.38 ± 0.81
400	0.001	15	11.94 ± 0.62	88.59 ± 0.07
400	0.01	15	11.15 ± 3.20	87.78 ± 0.09
400	0.1	15	19.84 ± 4.22	88.12 ± 0.46
1000	0.001	5	10.73 ± 0.57	91.77 ± 0.12
1000	0.01	5	11.98 ± 4.44	91.34 ± 0.45
1000	0.1	5	31.91 ± 3.73	92.16 ± 0.85
1000	0.05	5	17.58 ± 1.45	91.12 ± 0.73
1000	0.2	5	14.92 ± 0.88	91.62 ± 0.22
1000	0.001	15	11.88 ± 1.09	91.65 ± 0.36
1000	0.01	15	13.71 ± 5.71	91.31 ± 0.69
1000	0.1	15	20.22 ± 2.01	89.05 ± 0.34
1000	0.05	15	25.99 ± 7.68	91.06 ± 0.71
1000	0.2	15	20.81 ± 2.61	91.86 ± 0.61

Results: Training With Fault Correction



(a) Network trained with SA1 fault and no correction



(b) Network trained using proposed STDP-DIP framework

Results: Energy Saved

- Relative Energy reduction

$$E_{relative} = \left(\frac{V_{low}}{V_{nom}} \right)^2 = \left(\frac{1.00V}{1.30V} \right)^2 \approx 59.17\%$$

V_{low} : Undervolted potential

V_{nom} : Nominal potential

- 40.83% reduction by scaling supply voltage down to 1.00V
- 20.4% to 30.6% total energy reduction
 - Assumption: Memory energy 50% to 75%

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Conclusion

- Identified impact of SA1 memory faults on On-Chip STDP learning
- Developed STDP-DIP to improve accuracy
 - Restored >91% accuracy while saving 30.6% system energy
- Future Work
 - Explore different learning approach
 - Explore mixed-fault scenarios (SA0 + SA1)

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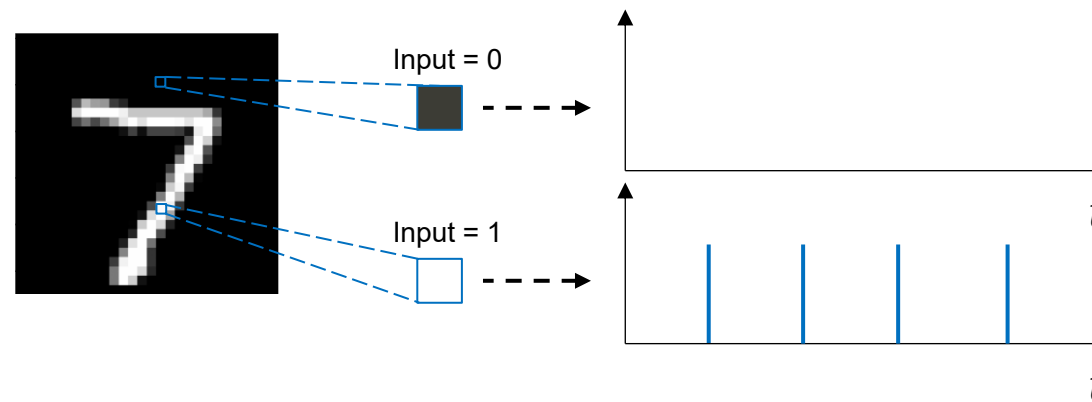
Thank you for your attention!

Details

- The DRAM cell is implemented using HSPICE
- Capacitance: 25 fF, Refreshing frequency: 15.625 Hz ^[9]
- Monte Carlo simulation is performed with Gaussian Variation on voltage
- Temperature is varied and its relation to the error percentage is found
- Ran training and testing of Neural Network, identified fault insertion parameters and location
- Faults have been inserted during training

Rate Coding

- Converts raw sensor data into spike trains and processed by an SNN
- High pixel intensity = High spike frequency
- Transmitting 1-bit spikes is more efficient



Spikes generated from MNIST image